ONCOLOGY · NEET-SS / FET / INI-SS

Bone-forming tumours

1. Overview and general principles

- **Bone-forming (osteogenic) tumours** are defined by tumour cells that **produce osteoid or bone directly**.
- **WHO 2020 classification of osteogenic tumours:**
  - **Benign:** osteoma, osteoid osteoma.
  - **Intermediate (locally aggressive):** osteoblastoma.
  - **Malignant:** osteosarcoma and its subtypes (conventional, telangiectatic, small cell, low-grade central, parosteal, periosteal, high-grade surface, secondary).
- **Osteosarcoma is the commonest primary malignant bone tumour** (excluding myeloma, which is a marrow neoplasm). Chondrosarcoma is second, Ewing sarcoma third (second in children).
- **Clinical age clue:** under 5 years, a destructive bone lesion is usually metastatic neuroblastoma or LCH; 10-25 years osteosarcoma or Ewing; over 40 years metastasis or myeloma first, then chondrosarcoma.
- **Investigation sequence for a suspicious lesion:** plain radiographs in two planes, **MRI of the whole bone** (extent, soft tissue mass, skip lesions, neurovascular relations), **CT chest** (lung metastases), bone scan or **PET-CT** (bone metastases), then **biopsy last**, at the treating sarcoma centre.
- **Biopsy rules (Mankin):** core needle preferred; tract **in line with the definitive incision** so it can be excised en bloc; go **through a single muscle compartment**, not between intermuscular planes; avoid neurovascular bundles and joints; meticulous haemostasis; drains brought out in line with the incision. Mankin showed that biopsies

done outside specialist centres caused significantly more errors and changes in treatment, including unnecessary amputation.

Surgical margins (Enneking)

Margin	Plane of dissection	Result for high-grade sarcoma
Intralesional	Through the tumour	Gross residual tumour; nearly always recurs
Marginal	Through the reactive zone (pseudocapsule)	Leaves satellite lesions; high recurrence
Wide	Through normal tissue with a cuff around the reactive zone	Standard goal for limb salvage
Radical	Entire anatomical compartment removed	Rarely needed now with chemotherapy

Enneking (MSTS) staging

Stage	Meaning
Benign 1 (latent)	Inactive, well-marginated, often heals spontaneously (e.g. NOF)
Benign 2 (active)	Growing, contained within bone (e.g. most osteoid osteomas)
Benign 3 (aggressive)	Breaches cortex, may metastasise rarely (e.g. osteoblastoma, GCT)
IA / IB	Low grade, intracompartmental (A) or extracompartmental (B)
IIA / IIB	High grade, intra- or extracompartmental. **Most osteosarcomas are IIB**
III	Any grade with distant metastases

- **AJCC 8th edition (appendicular skeleton):** T1 is 8 cm or less, T2 over 8 cm, T3 discontinuous tumours (skip lesions) in the primary bone; M1a lung only, M1b other distant sites. Stage IV = metastatic (IVA lung only; IVB other sites or nodes).

2. Osteoma and bone island

- **Osteoma:** dense lamellar bone on the surface of the **skull, facial bones and paranasal sinuses** (frontal sinus commonest). Usually asymptomatic.
- **Gardner syndrome:** multiple osteomas + **familial adenomatous polyposis** (APC gene, chromosome 5q) + desmoid tumours and epidermoid cysts. Colonic polyps have near 100% malignant potential, so osteomas of the jaw should prompt colonoscopy.
- **Bone island (enostosis):** intramedullary compact bone; **"thorny" radiating spicules (brush border)** blending into surrounding trabeculae; commonly pelvis and proximal femur.
  - Usually cold or mildly warm on bone scan, which helps distinguish it from **osteoblastic metastasis** (hot).
  - Over 2 cm = **giant bone island**. Growth over time warrants biopsy only if behaviour is atypical.
- **Osteopoikilosis:** multiple bone islands around joints (LEMD3 gene), asymptomatic; **Buschke-Ollendorff** when combined with skin lesions.

3. Osteoid osteoma

Epidemiology and pathology

- About 10-12% of benign bone tumours. **Age 5-25 years**; **male to female about 2-3:1**.
- **Sites: proximal femur (femoral neck and intertrochanteric region) is the commonest**, then tibia; about 10% in the spine (**posterior elements**, lumbar commonest); hand and foot.

- **Nidus under 1.5 cm** (some texts 2 cm): interlacing osteoid trabeculae lined by osteoblasts in vascular fibrous stroma, surrounded by **reactive sclerosis**. Not malignant; no malignant transformation.
- Pain mediated by **prostaglandin E2 and COX-2** production in the nidus; the nidus contains **unmyelinated nerve fibres**.
- **FOS gene rearrangements** are now recognised in both osteoid osteoma and osteoblastoma.

Clinical features

- **Night pain relieved dramatically by aspirin or NSAIDs** (classic; about 75%).
- **Spine: painful scoliosis** (stiff, non-structural), the **lesion is on the concave side at the apex** of the curve; paravertebral spasm. Early removal (within about 15 months) allows the curve to resolve.
- **Intra-articular (hip, elbow, ankle):** synovitis, effusion, stiffness, little sclerosis; mimics inflammatory arthritis or infection; delayed diagnosis common.
- **Near the physis in children:** overgrowth and limb length discrepancy; muscle wasting.

Imaging

- **Radiograph:** small **radiolucent nidus with dense surrounding sclerosis** and cortical thickening (cortical type commonest); the nidus may be hidden by sclerosis.
- **CT (thin slices) is the investigation of choice to locate the nidus**; nidus may show central mineralisation (target).
- **Bone scan:** intense uptake, **"double-density sign"** (hot nidus within a less hot reactive zone).
- **MRI:** marked marrow and soft tissue oedema may mislead (mimics infection or tumour); dynamic contrast shows early arterial enhancement of the nidus.

Differential diagnosis

- **Brodie abscess** (serpiginous lucent tract, often crosses physis, no nidus calcification), **stress fracture** (linear lucency perpendicular to cortex), osteoblastoma (larger), chronic osteomyelitis, intracortical haemangioma.

Management

Option	Indication / notes
NSAIDs	Many lesions burn out in 2-4 years (natural history); acceptable if tolerated
CT-guided radiofrequency ablation (RFA)	**Treatment of choice** for most accessible lesions; primary success about 90%, recurrence about 5-10%, can be repeated. Heats nidus to about 90 degrees C for 4-6 min
Surgical excision (en bloc or burr-down)	Lesion near skin (burn risk), near nerve roots or spinal cord (relative caution with RFA), failed RFA, need for histology
Spine	Excision or RFA with neural protection; early treatment to prevent structural scoliosis

- Recurrence of pain after treatment means incomplete nidus removal.

> Exam pearl: Osteoid osteoma = nidus under 1.5 cm, night pain relieved by NSAIDs (PGE2/COX-2), proximal femur commonest, CT best to find the nidus, "double-density" on bone scan, painful scoliosis with the lesion on the CONCAVE side at the apex, RFA is the treatment of choice.

4. Osteoblastoma

- **Over 1.5-2 cm**, histologically similar to osteoid osteoma but **progressively growing** and **locally aggressive** (Enneking benign 3 in many cases).
- **Age 10-30**, male predominance.
- **Spine and sacrum are the commonest sites (about 40%)**, again in the **posterior elements**; also long bones (metadiaphysis) and talus.
- Dull ache, **less night pain and poor response to NSAIDs**; may cause **radiculopathy or cord compression**, and painful scoliosis.
- **Imaging:** expansile lytic lesion with variable mineralisation and a thin rim of reactive bone; **secondary ABC change** is common; soft tissue extension.
- **Differential: osteoblastoma-like osteosarcoma** (permeative margin, infiltration of host trabeculae), ABC, GCT.
- **Treatment: extended curettage with adjuvant (high-speed burr, phenol, cryotherapy) and grafting**, or **en bloc resection** for aggressive and spinal lesions (with stabilisation). Recurrence after curettage about 10-20%. Radiotherapy is avoided.
- **Epithelioid ("aggressive") osteoblastoma** has a higher recurrence risk.

5. Conventional osteosarcoma

Epidemiology and risk factors

- **Bimodal:** peak in **adolescence (10-20 years, growth spurt)** and a second peak **over 60 years** (secondary: Paget disease, irradiation). Slight male predominance.
- **Sites:** metaphysis of long bones with the fastest growth: **distal femur > proximal tibia > proximal humerus** (about 50% around the knee). Axial (pelvis, spine) lesions in older patients carry a worse prognosis.

- **Genetic associations:**
  - **Hereditary retinoblastoma (RB1, chromosome 13q14)**: large increase in risk, also in irradiated fields.
  - **Li-Fraumeni syndrome (TP53, 17p13)**: osteosarcoma, breast cancer, brain tumours, leukaemia, adrenocortical carcinoma.
  - **Rothmund-Thomson syndrome (RECQL4)**: poikiloderma, radial ray defects, osteosarcoma.
  - Werner (WRN) and Bloom (BLM) syndromes (RecQ helicases).
- **Secondary osteosarcoma:**
  - **Paget sarcoma:** in about 1% of Paget disease (higher in polyostotic disease); pelvis, femur, humerus; new pain or swelling in pagetic bone; **very poor prognosis** (5-year survival under 10-15%).
  - **Post-radiation sarcoma (Cahan criteria):** arises in the radiation field, latency of **over 3-4 years** (median about 10-15 years), histologically different from the original lesion. Osteosarcoma is the commonest radiation-induced bone sarcoma.
  - Others: fibrous dysplasia, bone infarct, chronic osteomyelitis (rare).

Pathology

- **Diagnostic criterion: malignant spindle (pleomorphic) cells producing osteoid directly**. Amount of osteoid can be small.
- High grade; histological subtypes of conventional OS: **osteoblastic (commonest), chondroblastic, fibroblastic**; subtype does not change treatment.
- Grows through the cortex and lifts periosteum; spreads within the medulla (**skip metastases** in about 5-10% of patients, i.e. discontinuous foci in the same or transarticular bone) and **haematogenously** (lymph node spread rare).
- **Lungs are the commonest site of metastasis** (about 80-90% of metastatic cases), then bone. About 15-20% have detectable

metastases at presentation, but about 80% have micrometastases, which is why surgery alone gave only 10-20% survival before chemotherapy.

Clinical features

- **Pain** (initially activity-related, then constant and at night), swelling, warmth, restricted joint movement; pathological fracture in about 5-10%.
- Often attributed to sports injury, causing diagnostic delay.
- **Serum alkaline phosphatase and LDH:** raised in about half; **high LDH or ALP indicates worse prognosis**, and ALP is used to monitor relapse.

Radiology

- **Metaphyseal, poorly defined, permeative, mixed lytic-sclerotic** lesion with a wide zone of transition.
- **Cloud-like osteoid matrix** in the soft tissue mass.
- **Periosteal reactions: Codman triangle** (elevated periosteum at the edge; seen in any aggressive lesion, not specific), **sunburst** (spiculated radiating) pattern, interrupted lamellar reaction.
- **MRI:** extent of marrow involvement and **skip lesions** (image the **whole bone and adjacent joint**), physeal and epiphyseal involvement, joint invasion, neurovascular encasement - all decide whether limb salvage is possible.
- **CT chest** for lung nodules; bone scintigraphy or **PET-CT** for skeletal metastases.

Prognostic factors

Factor	Unfavourable
Metastases at diagnosis	**Most important adverse factor** (5-year survival about 20-30%; better if lung only and resectable)

Factor	Unfavourable
Histological response to neoadjuvant chemotherapy (Huvos)	**Less than 90% tumour necrosis = poor responder**
Site	Axial / pelvic tumours
Size	Large volume tumours (T2)
Surgical margin	Inadequate margin raises local recurrence
Laboratory	Raised LDH and ALP
Skip lesions	Behave like metastatic disease

Huvos (Rosen-Huvos) grading of chemotherapy response

Grade	Necrosis	Interpretation
I	Little or no effect (under 50%)	Poor responder
II	50-89%	Poor responder
III	90-99%	**Good responder**
IV	100% (no viable tumour)	**Good responder**

Management

1. **Neoadjuvant (pre-operative) chemotherapy** for about 10 weeks: **MAP = high-dose Methotrexate (with leucovorin/folinic acid rescue), doxorubicin (Adriamycin) and cisPlatin**. This was developed from the **Rosen T-10 protocol** at Memorial Sloan Kettering.
2. **Restaging** with MRI and CT chest.
3. **Surgery with wide margins**: limb salvage in about 85-90% of patients today.
4. **Adjuvant (post-operative) chemotherapy** continuing MAP (total treatment about 29-30 weeks); the resected specimen gives the

Huvos response grade.

5. **Pulmonary metastasectomy** for resectable lung metastases (at presentation or relapse) improves survival; it is a standard part of treatment.

- **Chemotherapy toxicities:** methotrexate (renal failure, mucositis; monitor levels, glucarpidase if delayed clearance), doxorubicin (**cumulative dose-related cardiomyopathy**; dexrazoxane is cardioprotective), cisplatin (**ototoxicity**, nephrotoxicity, neuropathy), ifosfamide (haemorrhagic cystitis, prevented by **mesna**; Fanconi-type nephrotoxicity).
- **Radiotherapy:** osteosarcoma is **relatively radioresistant**; used only for unresectable tumours (spine, pelvis, skull) or palliation, often as proton or carbon-ion therapy.
- **Survival:** localised extremity disease with modern treatment about **65-70% at 5 years**; metastatic about 20-30%. Survival has not improved substantially since the 1980s.

Landmark trials and evidence

- **MIOS (Multi-Institutional Osteosarcoma Study, Link et al., NEJM 1986)**: randomised trial showing that **adjuvant chemotherapy dramatically improves relapse-free survival** (about 66% versus 17% at 2 years) compared with surgery alone. Established chemotherapy as standard.
- **Simon et al. (MSTS, JBJS 1986)**: **limb salvage for distal femoral osteosarcoma gave survival equal to amputation** when wide margins were obtained.
- **EURAMOS-1 (published 2015-2016)**: largest osteosarcoma trial. In **poor responders**, adding **ifosfamide and etoposide (MAPIE)** to MAP **did not improve event-free survival** and increased toxicity. In **good responders**, adding **pegylated interferon alfa-2b** to MAP gave no significant benefit. Conclusion: **MAP remains the standard**

for all, and post-operative chemotherapy should not be changed on the basis of poor response.

- **INT-0133 (Children's Oncology Group)**: **mifamurtide (MTP-PE, liposomal muramyl tripeptide)**, an immune macrophage activator, improved overall survival when added to chemotherapy (results debated); **approved in Europe, not by the US FDA**.
- Targeted therapy: multi-kinase inhibitors (regorafenib, cabozantinib) show modest activity in relapsed disease only.

Surgical options

- **Limb salvage (contraindications are relative):** major neurovascular encasement that cannot be reconstructed, extensive soft tissue contamination (e.g. poorly placed biopsy, displaced pathological fracture through tumour), infection, very young child with large predicted leg-length discrepancy (now often managed with growing prostheses).
- **Reconstruction after resection:**
  - **Modular endoprosthesis (megaprosthesis)**: commonest; rotating-hinge knee for distal femur or proximal tibia.
  - **Expandable (growing) prosthesis** in skeletally immature children: non-invasive magnetic lengthening types avoid repeated surgery.
  - **Allograft** (osteoarticular or intercalary), **allograft-prosthesis composite**, vascularised fibula, recycled autograft (irradiated, frozen in liquid nitrogen or pasteurised).
  - **Rotationplasty (Van Nes)**: for distal femoral tumours in young children; the distal leg is rotated **180 degrees** and attached to the femur so that the **ankle functions as a knee** (plantarflexion = knee extension) and the foot fits a below-knee type prosthesis. Durable, active, no prosthetic loosening; cosmetic appearance is the main drawback.

  - **Amputation:** above-knee or hip disarticulation when limb salvage cannot achieve wide margins; same survival as limb salvage.
- **Proximal tibial resection:** the **extensor mechanism must be reconstructed** (medial gastrocnemius rotational flap with patellar tendon reattachment), and the gastrocnemius flap also covers the implant, reducing infection.
- **Endoprosthetic failure (Henderson classification):** type 1 **soft tissue failure** (instability, tendon failure), type 2 **aseptic loosening**, type 3 **structural failure** (implant or periprosthetic fracture), type 4 **infection**, type 5 **tumour progression**. Infection is the most feared and is particularly common after proximal tibial and pelvic reconstruction.
- **Local recurrence** after limb salvage is about 5-10% and carries a poor prognosis.
- **Follow-up:** CT chest and limb imaging every 3 months in years 1-2, then less frequently to 10 years; monitor late effects (cardiac, renal, hearing, fertility, second malignancy).

> Exam pearl: Conventional osteosarcoma: adolescent distal femoral metaphysis, mixed lytic-sclerotic with Codman triangle and sunburst. MAP neoadjuvant chemotherapy, wide resection, adjuvant MAP. Most important prognostic factor = metastases at diagnosis; next = Huvos necrosis (90% or more = good responder). EURAMOS-1: intensifying chemotherapy for poor responders does not help.

6. Subtypes of osteosarcoma

Subtype	Key features	Treatment and prognosis
Telangiectatic	Purely lytic, expansile, haemorrhagic; **fluid-fluid levels on MRI mimic ABC**; malignant cells in the septa (thick, nodular, enhancing septa on MRI)	As conventional (chemotherapy + wide resection); prognosis similar to conventional; pathological fracture common
Small cell	Small round cells producing lace-like osteoid; may be CD99 positive	**Main differential is Ewing sarcoma** (no EWSR1 translocation); treated as conventional; slightly worse prognosis
Low-grade central	Metaphysis of femur or tibia, adults 20-40; slowly growing; **mimics fibrous dysplasia**; **MDM2 and CDK4 amplification**	**Wide excision alone**; no chemotherapy; excellent prognosis unless it dedifferentiates
Parosteal	Commonest surface OS (about 5% of all OS); **posterior distal femoral metaphysis** (popliteal surface); women 20-40 years; dense, lobulated mass **"wrapped around" the bone**; **radiolucent cleavage line (string sign)** between tumour and cortex; **MDM2/CDK4 amplification**; low grade	**Wide excision alone**; 5-year survival about 90%; marginal excision recurs; **dedifferentiated parosteal OS** (about 15-25%) is high-grade and treated as conventional

Subtype	Key features	Treatment and prognosis
Periosteal	Surface of the **diaphysis of tibia or femur**, adolescents; **predominantly chondroblastic**, intermediate grade; saucerised cortex with **perpendicular "hair-on-end" spicules** and Codman triangle; medulla spared	Wide excision; **role of chemotherapy debated** (often given); prognosis better than conventional, worse than parosteal
High-grade surface	Rare, surface lesion but histologically like conventional	As conventional (chemotherapy + surgery)
Secondary (Paget, irradiation, infarct)	Older patients	Chemotherapy often poorly tolerated; poor prognosis
Extraskeletal osteosarcoma	Soft tissue of thigh in adults over 50	Treated like soft tissue sarcoma; poor prognosis

- **Parosteal versus myositis ossificans (most important differential):**
  - **Myositis ossificans: zonal pattern** with **mature bone at the periphery and immature centre**; history of trauma; matures over 6-8 weeks; separated from cortex.
  - **Parosteal osteosarcoma: densest centrally at the base**, less mature at the periphery (reverse zonal pattern); attached to cortex by a broad base.
  - Also differentiate from osteochondroma (corticomedullary continuity, cartilage cap) and **BPOP** (bizarre parosteal osteochondromatous proliferation, hands and feet).

Exam pearl: Parosteal OS = posterior distal femur, woman 20-40, "string sign", MDM2 amplification, low grade, wide excision only, about 90% survival. Periosteal OS = tibial diaphysis, chondroblastic, hair-on-end. Telangiectatic OS = fluid-fluid levels like ABC but with thick nodular septa - never curette an "ABC" without histology.

7. Summary of radiological differentials for a bone-forming lesion

Lesion	Discriminating feature
Stress fracture	Linear periosteal/endosteal reaction, no soft tissue mass, history of activity
Exuberant callus	History of fracture, matures over weeks
Osteomyelitis	Sequestrum, cloaca, systemic features; can mimic Ewing or OS
Ewing sarcoma	Diaphyseal, onion-skin periosteal reaction, large soft tissue mass, no osteoid matrix
Osteoblastic metastasis	Older patient, multiple lesions (prostate, breast)
Paget disease	Coarsened trabeculae, cortical thickening, bone enlargement, starts at an end of the bone and advances (blade of grass / flame-shaped front in long bones)

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.